Performance Testing of an External Gear Pump

Displacement, slip, volumetric and overall efficiency, and power losses of a positive-displacement pump

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Team	Group A4: Luis Ramirez, Macklin Horner, Zachary Klein, Larimel Rosario, John Wahba	**Equipment**	GUNT HM 365 drive and brake unit with HM 365.20 gear pump module, hydraulic oil (SG 0.88), pressure, flow, torque and speed sensors

PROJECT SUMMARY

An external gear pump pumping hydraulic oil was tested at 800, 1200, and 1600 rpm. At each speed the delivery valve was closed in steps until the discharge pressure reached about 95 bar, and the HM 365.20 software recorded torque, speed, suction and delivery pressure, flow rate, and oil temperature. The delivered flow stayed almost constant with pressure, as expected for a positive-displacement pump: extrapolating to zero head gave a displacement of about 2.38 cm³ per revolution, and leakage (slip) never exceeded 2.5% of the flow. Overall efficiency rose with pressure to a maximum of about 74%, while the friction loss stayed nearly constant at each speed and grew with speed, from about 100 W at 800 rpm to about 210 W at 1600 rpm.

1. Introduction

Gear pumps are positive-displacement machines used wherever a steady flow must be delivered against high pressure, such as hydraulic power units, lubrication systems, and fuel and chemical transfer. Two meshing gears trap fluid in the spaces between their teeth and the casing and carry it from the suction side to the discharge side, so each revolution moves a fixed volume. In contrast to a centrifugal pump, the flow barely depends on the head; the only loss of flow is internal leakage, or slip, back across the gear faces and tips. The objective of this experiment was to measure the performance of a gear pump at three speeds and determine its displacement, slip, volumetric and overall efficiency, and the split between useful power and friction loss.

2. Theory

Table 1. Performance relations (oil SG = 0.88, γ = 8632.8 N/m³)

Quantity	Relation	Notes
Pump head	$H_p = (p_2 - p_1)/\gamma$	Delivery minus suction pressure, in m of oil
Theoretical capacity	$Q_t = Q$ at $H_p = 0$	Intercept of the linear Q–H_p fit; displacement $V_d = Q_t/n$
Slip	$Q_s = Q_t - Q$	Internal leakage from discharge back to suction
Volumetric efficiency	$\eta_v = Q/Q_t$	Fraction of the displaced volume delivered
Hydraulic power	$E_h = Q(p_2 - p_1)$	Useful power given to the oil
Brake power	$E_b = M\omega$	Shaft torque times angular speed
Friction loss / efficiency	$E_f = E_b - E_h$, $\eta = E_h/E_b$	Mechanical and viscous losses; overall efficiency

Because the shaft must overcome a roughly fixed amount of bearing, seal, and viscous friction at a given speed, the overall efficiency is low at small pressure rises and improves as the hydraulic power grows. The net positive suction head, $NPSH = (p_1 + p_{atm} - p_v)/\gamma + V_s^2/2g$, was also checked to confirm the pump was not cavitating.

3. Procedure

The pump was started with the delivery valve fully open and brought to 800 rpm. The valve was then closed in small steps, and at each step the readings were allowed to settle before the software logged torque, speed, suction and delivery pressure, flow rate, and oil temperature. Closing continued until the delivery pressure reached about 95 bar, safely below the 100 bar relief valve setting. The run was repeated at 1200 and 1600 rpm, giving 17–20 operating points per speed. Oil temperature stayed between 22.5 and 23.7 °C, so viscosity changes between runs were negligible.



Figure 1. Gear pump test stand in the lab

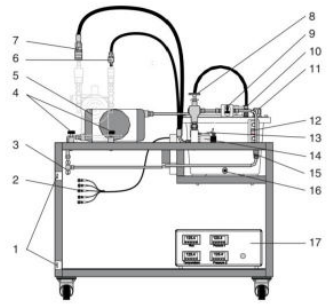


Figure 2. HM 365.20 module layout (lab manual)

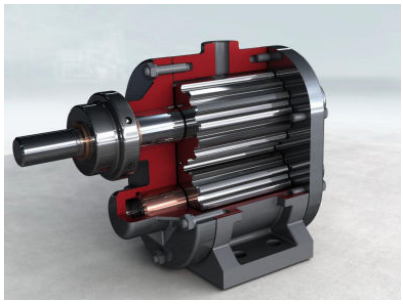


Figure 3. Cutaway of an external gear pump

4. Results

Delivered flow and slip

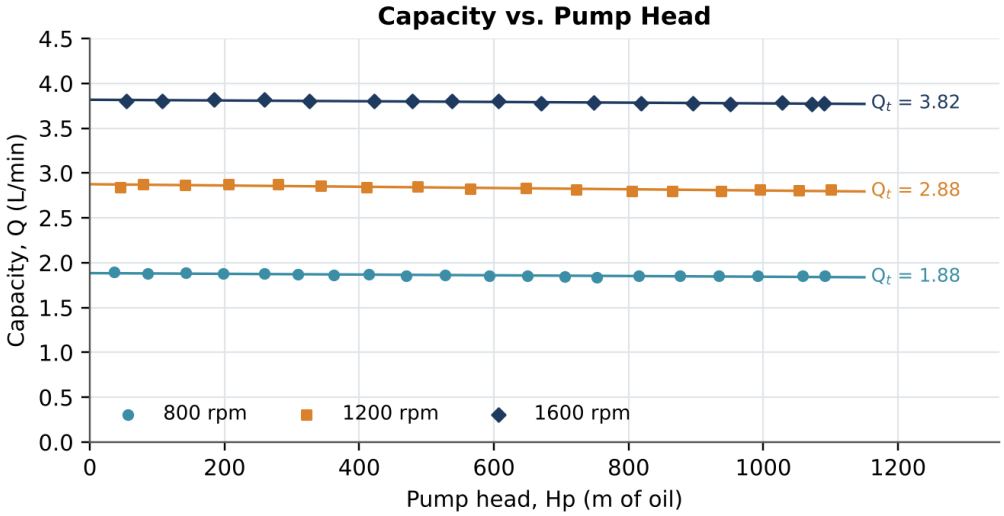


Figure 4. Capacity vs. pump head at three speeds; lines are linear fits extrapolated to Hp = 0 to find Q_t (L/min).

Table 2. Summary of results at each speed

Speed (rpm)	Q_t (L/min)	V_d (cm³/rev)	Max p_2 (bar)	Max slip (L/min)	Min η_v (%)	Max η (%)	E_f (W)
800	1.884	2.369	94.2	0.043	97.7	73.9	100–105
1200	2.875	2.400	95.0	0.073	97.5	74.4	140–163
1600	3.817	2.384	94.1	0.048	98.8	72.6	199–231

Efficiency and power

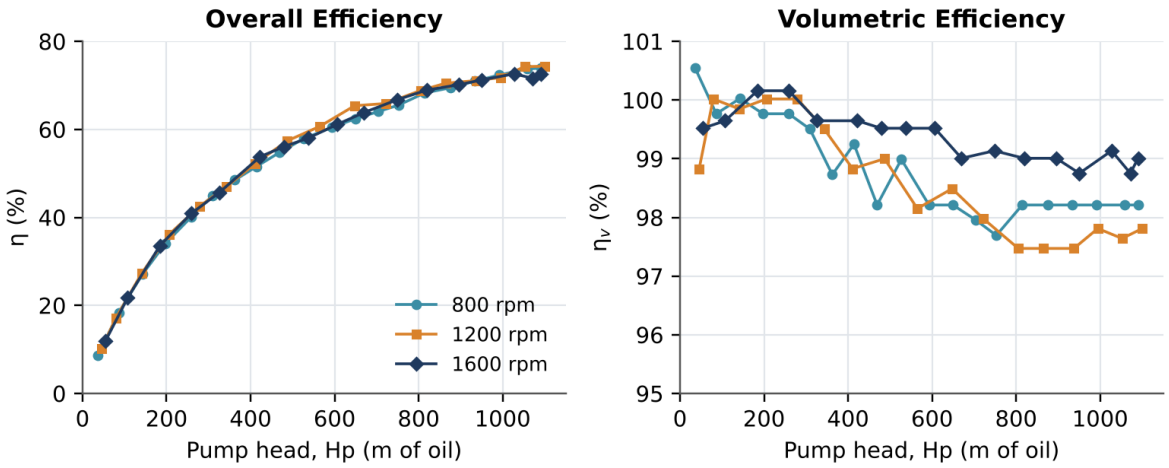


Figure 5. Overall and volumetric efficiency vs. pump head.

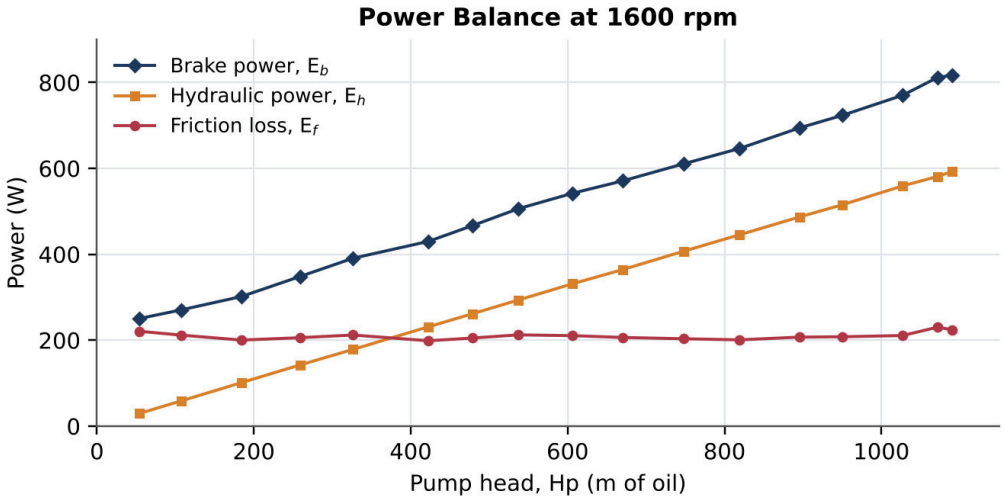


Figure 6. Brake power, hydraulic power, and friction loss vs. pump head at 1600 rpm.

5. Discussion

Positive-displacement behavior. At every speed the flow stayed nearly flat as the head rose from about 36 m to over 1000 m of oil, which is the defining feature of a gear pump. The three extrapolated displacements, 2.37, 2.40, and 2.38 cm³/rev, agree within about 1%, so the theoretical capacity scales directly with speed. Slip grew slightly with pressure but stayed below 0.08 L/min, and the volumetric efficiency never dropped below 97.5%, meaning the gear clearances were tight and the oil viscosity was high enough to limit leakage.

Overall efficiency and losses. Overall efficiency was low at small pressure rises (9–12%) and climbed steadily to about 73–74% near 95 bar at all three speeds (72.6% at 1600 rpm). The reason is visible in the power balance: the friction loss stayed almost constant along each run (about 100–105 W at 800 rpm, 140–163 W at 1200 rpm, and 199–231 W at 1600 rpm), while hydraulic power increased in proportion to pressure. The loss therefore becomes a smaller share of the shaft power as pressure rises. Because the friction loss roughly doubles from 800 to 1600 rpm while the hydraulic power also doubles, the peak efficiency is similar at every speed.

Suction conditions and accuracy. The suction pressure stayed within a few hundredths of a bar of atmospheric, giving NPSH ≈ 11.5 m at every point, so cavitation was not a concern. The small negative slip values at a few low-pressure points (flow slightly above Q_t) come from the resolution of the flow sensor and the scatter around the linear fit, not from a physical gain in flow.

6. Conclusion

The external gear pump delivered a nearly constant flow of 1.88, 2.88, and 3.82 L/min at 800, 1200, and 1600 rpm, corresponding to a displacement of about 2.38 cm³/rev. Slip stayed below 2.5% of the flow, giving volumetric efficiencies above 97.5%. Overall efficiency rose with discharge pressure to about 74%, limited by a friction loss that was almost independent of pressure but grew with speed. These results confirm the expected behavior of a positive-displacement pump and show that it performs best when operated near its rated pressure.

References

New Jersey Institute of Technology. (n.d.). *Performance test on gear pump* [ME 405 laboratory manual]. Department of Mechanical and Industrial Engineering.

Munson, B. R., Young, D. F., & Okiishi, T. H. (2006). *Fundamentals of fluid mechanics* (5th ed.). Wiley.